

OFFICIAL AGROSYNC TECHNICAL SPECIFICATION

System Design and Architecture Document

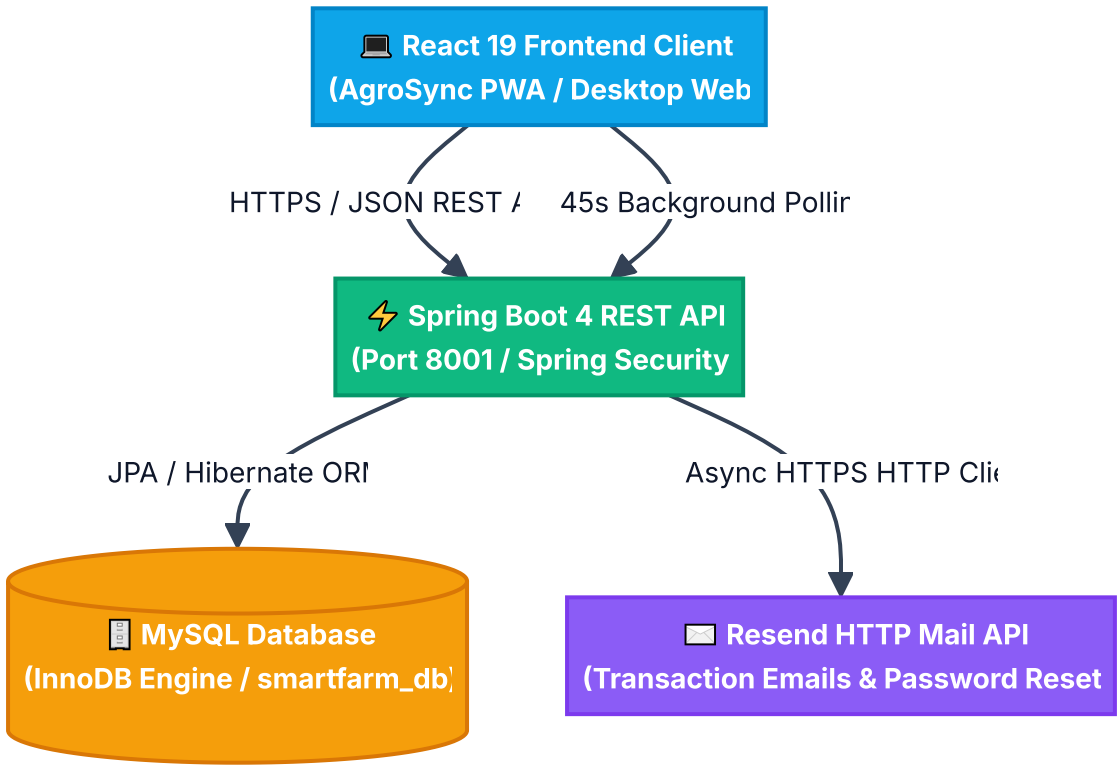
AgroSync Farm Management System

Version: 3.0

SDLC Stage: Technical Architecture & Status: Approved
Production Blueprint

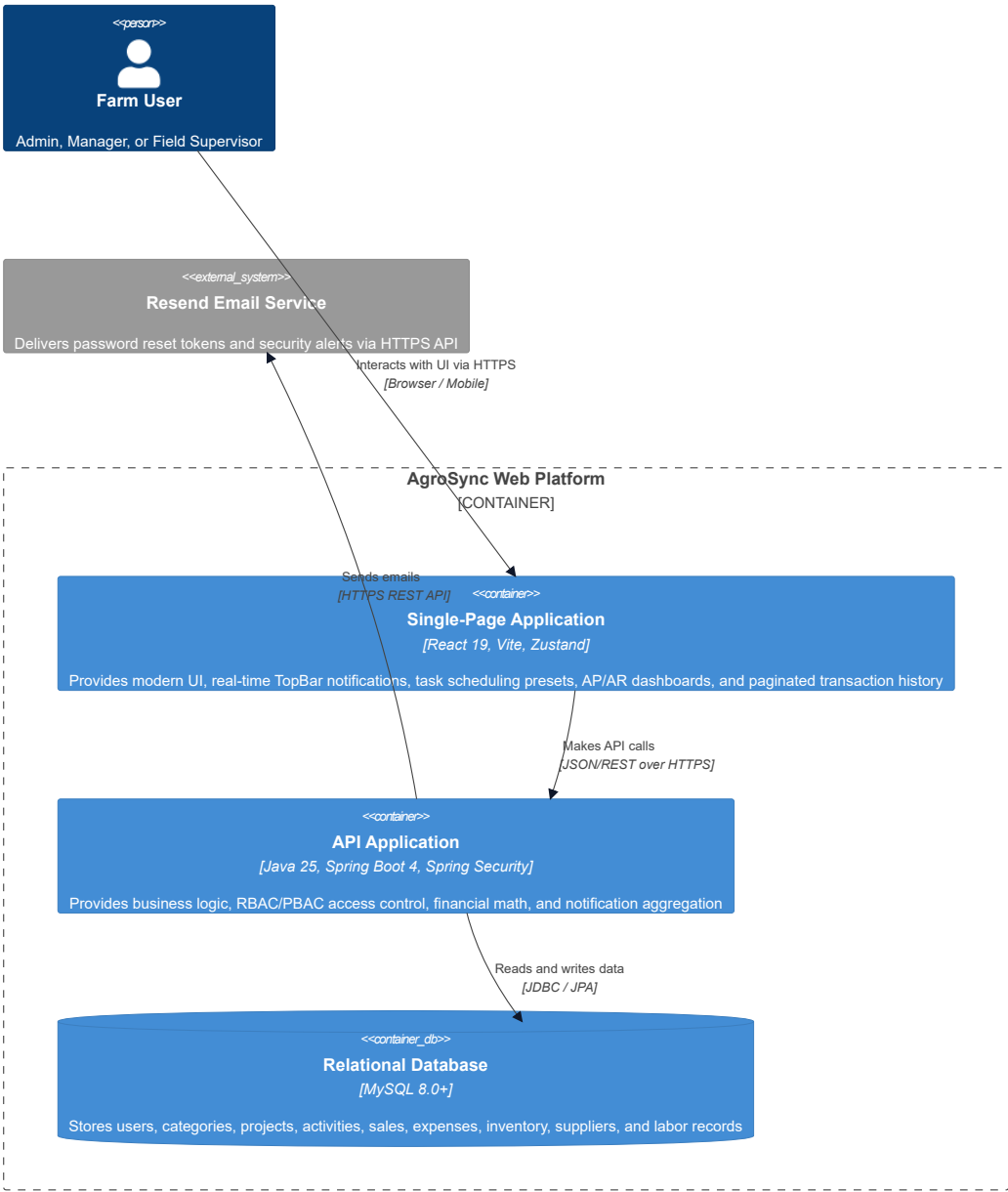
1. Executive Summary & Architectural Overview

AgroSync is built on a decoupled, cloud-native architecture combining a high-performance **Spring Boot 4 (Java 25)** micro-monolith backend with a reactive **React 19 (Vite)** single-page web client.

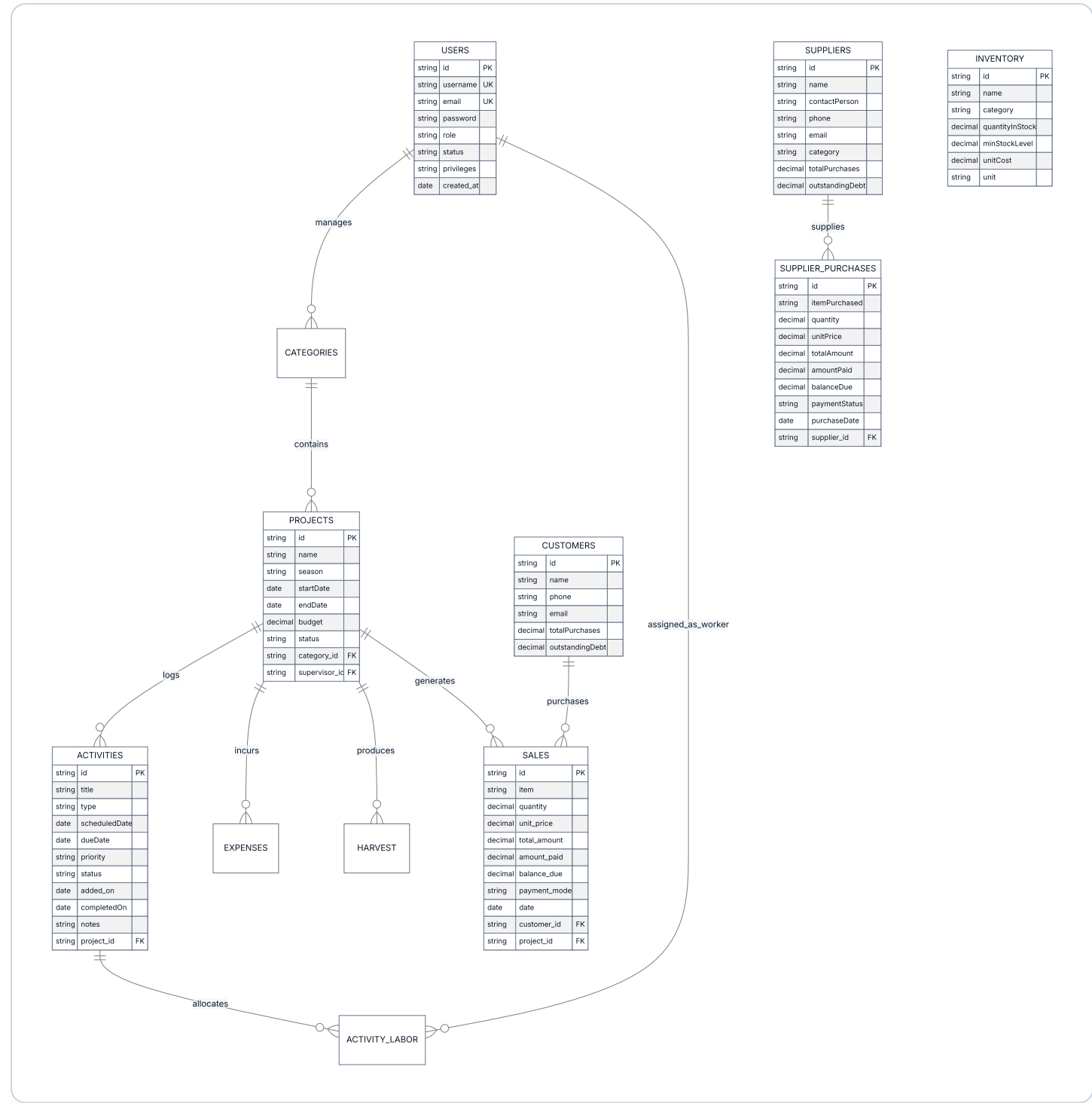


2. C4 Container Diagram

Container Diagram for AgroSync System



3. Entity-Relationship Data Model (ERD)



4. REST API Endpoint Catalog

4.1 Notifications & Real-Time Alerts

Method	Endpoint	Role Guard	Description
`GET`	`/notifications`	All Authenticated	Returns real-time pending approvals, low stock, upcoming/overdue tasks, and debt alerts

4.2 Activities & Task Scheduler

Method	Endpoint	Role Guard	Description
`POST`	`/activities/record`	Admin, Manager, Supervisor (`CAN_LOG_ACTIVITIES`)	Creates a new farm task / activity with scheduled date and priority
`PUT`	`/activities/{id}`	Admin, Manager	Updates task details, scheduled date, due date, priority, and status
`PATCH`	`/activities/{id}/status?status=COMPLETED`	Admin, Manager, Supervisor	Toggles task status between `SCHEDULED` and `COMPLETED`
`DELETE`	`/activities/{id}`	Admin, Manager	Deletes activity record
`GET`	`/activities/{id}/labor`	All Authenticated	Retrieves labor roster and wage allocations for this task
`POST`	`/activities/{id}/labor`	Admin, Manager, Supervisor	Assigns verified workers with wages to this activity

4.3 Dashboard & Financial Ledger

Method	Endpoint	Role Guard	Description
`GET`	`/dashboard/summary`	All Authenticated (Scoped by Role)	Returns executive KPIs (Revenue Received, Customer Debt, Supplier Debt, Net Value, Sales Trends, Recent Transactions)

4.4 Suppliers (Accounts Payable)

Method	Endpoint	Role Guard	Description
`GET`	`/suppliers`	Admin, Manager	Lists all suppliers and total purchase liabilities
`POST`	`/suppliers`	Admin, Manager	Registers a new supplier
`POST`	`/suppliers/purchases`	Admin, Manager	Records supply purchase with AP ledger tracking

4.5 Customers (Accounts Receivable)

Method	Endpoint	Role Guard	Description
`GET`	`/customers`	Admin, Manager	Lists all customers with outstanding debt balances
`POST`	`/customers`	Admin, Manager, Supervisor	Registers customer / credit profile

5. Unified Dashboard & Transaction History Data Architecture

5.1 Transaction Consolidation & Ingestion

The dashboard consolidates incoming revenue operations (Produce Sales) and outgoing input procurement (Supplier Purchases) into a singular, normalized data stream:

```

{
  id: String,
  type: 'SALE' | 'SUPPLY',
  date: 'YYYY-MM-DD',
  partyName: String, // Customer name or Supplier name
  item: String,       // Crop/produce sold or input purchased
  amount: Number,     // Total value
  paid: Number,       // Inflow or Outflow amount settled
  balance: Number,    // Remaining debt balance
  paymentStatus: 'PAID' | 'PARTIAL' | 'UNPAID',
  paymentMode: String
}

```

5.2 Deterministic Ordering & Pagination State Machine

1. **Filtering:** Applies search string against `partyName`, `item`, `id`, and payment mode, combined with type filter (`ALL`, `SALES`, `SUPPLIES`).

2. **Latest-First Sorting:**

```
````javascript
```

```

const sorted = filtered.sort((a, b) => {

const dateA = a.date || '';
const dateB = b.date || '';

const dateDiff = dateB.localeCompare(dateA);

if (dateDiff !== 0) return dateDiff;

return String(b.id || '').localeCompare(String(a.id || ''));

});
...

```

3. **Fixed-Size Slicing ( `TX_PAGE_SIZE = 5` ):**

- `totalTxPages = Math.ceil(sorted.length / 5)`
- `paginatedTx = sorted.slice((page - 1) * 5, page * 5)`
- Auto-reset hook resets `page → 1` whenever search or category filter updates.

## 6. Security & Access Control Architecture

1. **Stateless Header-Based Verification:**

- `X-User-Id` : Identifies the authenticated actor.
- `X-User-Role` : Enforces `ADMIN`, `MANAGER`, or `SUPERVISOR` boundaries.

2. **Supervisor Isolation:**

- Supervisor requests automatically filter to records where `supervisor_id = current_user.id`.
- Financial aggregate amounts (e.g. Total farm revenue, global debts) are completely hidden for supervisors without the `CAN_VIEW_FINANCIALS` privilege.

3. **CORS & CSRF:**

- Configured in `SecurityConfig.java` to allow secure API requests from authorized frontend hosts while enforcing strict header policies.